

**DATA 3464: Fundamentals of Data Processing**

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# Signals and images

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# Topic overview

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- Introduction to signals
- Images as 2D signals
- File formats
- Signal and image processing (for machine learning)

## Resources used:

- Various textbooks from my undergrad
- [DSPguide.com](https://dspguide.com) seems like a pretty good resource

# What is a signal?

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*"A [continuous/discrete] signal is a function of independent variables that range over [a continuum/discrete] values" - Jerry L. Prince, Medical Imaging Signals and Systems*

- Common notation:  $x(t)$  for continuous,  $x[n]$  for discrete
- Signals are **discretized** by **sampling** at some fixed interval  $dt$
- The **sampling rate** is informed by the frequency content of the data:

$$f_s \geq 2f_{max}$$

(but in practice is much higher)

# Frequency content of a signal

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- A discrete time domain signal can be represented as:

$$x[n] = \sum_{k=0}^{N-1} \left[ a_k \cos \left( \frac{2\pi kn}{N} \right) + b_k \sin \left( \frac{2\pi kn}{N} \right) \right]$$

- Or, using Euler's formula  $e^{j\theta} = \cos \theta + j \sin \theta$ :

$$x[n] = \sum_{k=0}^{N-1} c_k e^{j \frac{2\pi kn}{N}}$$

where the complex coefficients  $c_k = a_k + jb_k$  and  $j = \sqrt{-1}$

# Fourier Transform

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- To figure out what the coefficients  $c_k$  are, we can use the **Discrete Fourier Transform (DFT)**:

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi \frac{k}{N} n}$$

where each element of  $X[k]$  is the coefficient  $c_k$  for frequency  $k$

- This can also be inverted to get back the original signal:

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j2\pi \frac{k}{N} n}$$

# Symmetry in the frequency domain

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- Since a real-valued signal in time is composed of both sine and cosine components, its DFT has **conjugate symmetry**

$$X[N - k] = X[k]^*$$

where  $*$  denotes the complex conjugate

- This means the **negative-frequency half** of the spectrum is redundant
- In practice, for real-valued data, we often only inspect:
  - **magnitude**:  $|X[k]|$  to see "how much" of each frequency is present
  - **phase**:  $\angle X[k]$  to see alignment/shift information

# Example signal: Audio

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- Once you think of a signal as being a weighted sum of frequency components, you can do some fun things with it
- We can extract information, downsample, remove noise, etc
- Example: a typical .wav file
  - Uncompressed
  - 16 bits per sample (bit depth)
  - 48 kHz sampling rate
  - mono (1 channel) or stereo (2 channels)

*What about .mp3? .ogg? I would use [ffmpeg](#) to convert to .wav*